LAB 2

PRELIMINARY SCANNING APPLICATIONS

NAME: Hiten Raj Singh

Roll No: 23, CSE C

Example Q.

```
#include <stdio.h>
#include <stdlib.h>
int main() {
    FILE *fa, *fb;
    char ca, cb;
    fa = fopen("in.c", "r");
    if (fa == NULL) {
        printf("Cannot open file \n");
        exit(0);
    }
    fb = fopen("out.c", "w");
    ca = getc(fa);
    while (ca != EOF) {
        if (ca == ' ') {
            putc(ca, fb);
            while (ca == ' ')
                ca = getc(fa);
        }
        if (ca == '/') {
            cb = getc(fa);
            if (cb == '/') {
                while (ca != '\n')
                    ca = getc(fa);
            }
            else if (cb == '*') {
                do {
                    while (ca != '*')
                        ca = getc(fa);
                    ca = getc(fa);
                } while (ca != '/');
            }
            else {
                putc(ca, fb);
```

```
                putc(cb, fb);
            }
        }
        else {
            putc(ca, fb);
        }

        ca = getc(fa);
    }
    fclose(fa);
    fclose(fb);
    return 0;
}
```
Screenshot:

```
6CSEC1@dbl-23:~/Documents/CD_230905152/L2$ cc -o abc demo.c
6CSEC1@dbl-23:~/Documents/CD_230905152/L2$ ./abc
6CSEC1@dbl-23:~/Documents/CD_230905152/L2$ cat in.c
//This is a single line comment
9/* *****This is a
******Multiline Comment
**** */
#include <stdio.h>
void main(){
        FILE *fopen(), *fp;
        int c ;
        fp = fopen( “prog.c”, “r” ); //Comment
        c = getc( fp ) ;
        while ( c!=EOF ){
                putchar( c );
                c = getc ( fp );
        }
        /*multiline
        comment */
        fclose(fp );
}
6CSEC1@dbl-23:~/Documents/CD_230905152/L2$ cat out.c
9
#include <stdio.h>
void main(){
        FILE *fopen(), *fp;
        int c ;
        fp = fopen( “prog.c”, “r” );    c = getc( fp ) ;
        while ( c!=EOF ){
                putchar( c );
                c = getc ( fp );
        }

        fclose(fp );
}
6CSEC1@dbl-23:~/Documents/CD_230905152/L2$
```

Q1.

```
#include <stdio.h>
#include <stdlib.h>
int main() {
    FILE *fa, *fb;
    char ca;
    fa = fopen("in1.c", "r");
    if (fa == NULL) {
        printf("Cannot open input file\n");
        exit(0);
```

```
    }
    fb = fopen("out1.c", "w+");
    ca = getc(fa);
    int in_qots = 0;
    while (ca != EOF) {
        if (ca == '\"') {
            in_qots++;
        }
        if (in_qots%2 == 0 && (ca == ' ' || ca == '\t')) {
            putc(' ', fb);
            while (ca == ' ' || ca == '\t' && ca != EOF) ca=getc(fa);
            continue;
        }
        putc(ca, fb);
        ca = getc(fa);
    }
    fclose(fa);
    fclose(fb);
    return 0;
}
```
Screenshot:

```
6CSEC1@dbl-23:~/Documents/CD_230905152/L2$ cc -o abc q1.c
6CSEC1@dbl-23:~/Documents/CD_230905152/L2$ ./abc
Processing complete.
6CSEC1@dbl-23:~/Documents/CD_230905152/L2$ cat in1.c
#include <stdio.h>

int     main() {
    int    a    =    10;      int    b =  20;
    printf("Hello World");
    return      0;
}

6CSEC1@dbl-23:~/Documents/CD_230905152/L2$ cat out1.c
#include <stdio.h>

int main() {
 int a = 10; int b = 20;
 printf("Hello World");
 return 0;
}

6CSEC1@dbl-23:~/Documents/CD_230905152/L2$
```

Q2.

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
//Preprocessor directives are those which start with # in the line of code
int main() {
    FILE *fa = fopen("in1.c", "r");
    if (fa == NULL) {
        printf("Cannot open file\n");
        exit(0);
    }
    FILE *fb = fopen("out2.c", "w+");
    char line[500];
    char c;
    int i = 0;
    while ((c = fgetc(fa)) != EOF) {
        line[i++] = c;
        if (c == '\n') {
            line[i] = '\0';
```

```c
            if (line[0] != '#') {
                fputs(line, fb);
            }
            i = 0;
        }
    }
    //Last Line- Edge Case as it might not have '\n' and EOF comes
    if (i > 0) {
        line[i] = '\0';
        if (line[0] != '#') {
            fputs(line, fb);
        }
    }
    fclose(fa);
    fclose(fb);
    return 0;
}
```

Note: If there is error after the # we need to check that:

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
int main() {
    FILE *fa = fopen("in2.c", "r");
    if (fa == NULL) {
        printf("Cannot open file\n");
        exit(0);
    }
    FILE *fb = fopen("out2.c", "w+");
    char line[500];
    int i = 0;
    char c;
    while ((c = fgetc(fa)) != EOF) {
        line[i++] = (char)c;
        if (c == '\n') {
            line[i] = '\0';
            if (strncmp(line, "#include", 8) != 0 && strncmp(line, "#define",
7) != 0) {
                for (int j = 0; j < i; j++) {
                    fputc(line[j], fb);
                }
            }
            i = 0;
        }
```

```
    }
    fclose(fa);
    fclose(fb);
    return 0;
}
```

Screenshot:

```
6CSEC1@dbl-23:~/Documents/CD_230905152/L2$ cc -o abc q2.c
6CSEC1@dbl-23:~/Documents/CD_230905152/L2$ ./abc
6CSEC1@dbl-23:~/Documents/CD_230905152/L2$ cat in2.c
#include <stdio.h>
#include <stdlib.h>

int main() {
 int a = 10; int b = 20;
 printf("Hello World");
 return 0;
}

6CSEC1@dbl-23:~/Documents/CD_230905152/L2$ cat out2.c



int main() {
 int a = 10; int b = 20;
 printf("Hello World");
 return 0;
}

6CSEC1@dbl-23:~/Documents/CD_230905152/L2$ 
```

Q3.

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <ctype.h>
const char *keywords[] = {
    "auto", "break", "case", "char", "const", "continue", "default", "do",
    "double", "else", "enum", "extern", "float", "for", "goto", "if",
    "int", "long", "register", "return", "short", "signed", "sizeof", "static",
```

```c
    "struct", "switch", "typedef", "union", "unsigned", "void", "volatile",
"while"
};

int is_key(char *buffer) {
    for (int i = 0; i < 32; i++) {
        if (strcmp(buffer, keywords[i]) == 0) {
            return 1;
        }
    }
    return 0;
}

int main() {
    FILE *fa, *fb;
    char ca;
    char buffer[50];
    int i = 0;
    int inside_qote = 0;

    fa = fopen("in3.c", "r");
    fb = fopen("out3.c", "w+");

    if (fa == NULL) {
        printf("Cannot open file\n");
        exit(0);
    }

    ca = getc(fa);
    while (ca != EOF) {
        if (isalnum(ca) || ca == '_') {
            buffer[i++] = ca;
        }
        else {
            if (i > 0) {
                buffer[i] = '\0';
                if (inside_qote%2==0 && is_key(buffer)) {
                    for (int j = 0; buffer[j]; j++) {
                        putc(buffer[j]-32, fb);
                    }
                } else {
                    fputs(buffer, fb);
                }
```

```c
                i = 0;
            }
            if (ca == '\"') {
                inside_qote++;
            }
            putc(ca, fb);
        }
        ca = getc(fa);
    }
    fclose(fa);
    fclose(fb);
    return 0;
}
```

Screenshot:

```
6CSEC1@dbl-23:~/Documents/CD_230905152/L2$ cc -o abc q3.c
6CSEC1@dbl-23:~/Documents/CD_230905152/L2$ ./abc
6CSEC1@dbl-23:~/Documents/CD_230905152/L2$ cat in3.c
#include <stdio.h>

int main() {
 int a = 10; int b = 20;
 printf("Hello World");
 return 0;
}

6CSEC1@dbl-23:~/Documents/CD_230905152/L2$ cat out3.c
#include <stdio.h>

INT main() {
 INT a = 10; INT b = 20;
 printf("Hello World");
 RETURN 0;
}

6CSEC1@dbl-23:~/Documents/CD_230905152/L2$
```